

Generator
Matrix G

$$\begin{bmatrix} \mathbf{I}_k & \\ & \mathbf{A}^T \end{bmatrix}$$

$k \times k$

$(n - k) \times k$

$\times$

Message Vector

$$\mathbf{x} = \begin{pmatrix} x_1 \\ \vdots \\ x_k \end{pmatrix}$$

$k \times 1$

$=$

$$\begin{bmatrix} \mathbf{x} \\ \mathbf{A}^T \mathbf{x} \end{bmatrix}$$

$k \times 1$

$(n - k) \times 1$

$\leftarrow$ Info

Systematic
Codeword
 $\mathbf{c} = G\mathbf{x}$

$\leftarrow$ Parity